PHASE 6

Python EDA & Root-Cause Analysis

From analysis-ready SQL data to exploratory insights in Pandas

B2B Travel Booking & Revenue Analytics

Complete Understanding Notes | Beginner-Friendly | Portfolio & Interview Ready

Purpose: understand what was done, why it was done, how it was validated, and how to explain it in an interview.

Prepared from the completed project work and verified analysis results.

Contents and Learning Map

Use this document in two ways: first as a learning summary, then as an interview revision sheet. Each section explains the concept, implementation, validation, and business meaning.

#	Section	What you should understand
1	Objective and workflow	Why Python was used after SQL and what questions it answered.
2	Database connection and loading	Safe PostgreSQL connection, query execution, and analysis grain.
3	Data quality and descriptive EDA	Shape, missingness, status mix, and booking-value distribution.
4	Outlier analysis	IQR method, business interpretation, and why outliers were not automatically deleted.
5	Trend and partner analysis	Monthly performance plus agent, destination, supplier, and hotel analysis.
6	Cancellation and correlation analysis	Refund drivers, cancellation reasons, and relationship diagnostics.
7	Exports, validation, and interview revision	What was exported, verified, and how to explain the phase.

Project rule: Do not memorize code alone. Be able to explain the grain of the data, why each step is needed, and how you verified the result.

1. Objective and End-to-End Workflow

Phase 6 uses Python as an analytical layer on top of the cleaned and currency-normalized PostgreSQL data. SQL created trusted datasets; Python was used to explore distributions, investigate unusual values, compare business segments, and generate analysis tables for further reporting.

Core idea: SQL prepared the trusted analytical layer. Python helped ask flexible follow-up questions and understand patterns before final dashboarding.

Primary analysis table

```
analytics.fact_bookings_fx  ->  119,877 analysis-ready bookings
```

The booking fact table is at booking grain: one valid row per booking. Monetary fields are already normalized to INR, so totals can be aggregated safely.

Questions answered in Python

- What does a normal booking look like, and how skewed is booking value?
- Are high-value bookings errors or plausible business transactions?
- Which months show the strongest and weakest GBV growth?
- Which agents, destinations, suppliers, and hotels contribute most?
- Which cancellation reasons create the largest operational and refund exposure?
- Which booking characteristics are associated with higher GBV?

2. PostgreSQL Connection and Loading

The project connected Jupyter/Pandas to PostgreSQL using SQLAlchemy and psycopg2. A connection URL object is safer than manually concatenating a password into a connection string, especially when the password contains special characters.

```
from sqlalchemy import create_engine, URL
import pandas as pd

url = URL.create(
    'postgresql+psycopg2',
    username='postgres',
    password='YOUR_PASSWORD',
    host='127.0.0.1',
    port=5432,
    database='travel_analytics'
)
engine = create_engine(url)

with engine.connect() as conn:
    print(conn.exec_driver_sql('SELECT 1').fetchone())
```

Expected connection check: (1,) confirms that Python can execute a query against PostgreSQL. Never commit a real password to GitHub or share it in screenshots.

```
query = 'SELECT * FROM analytics.fact_bookings_fx'
bookings = pd.read_sql(query, engine)
bookings.shape
```

Check	Verified result	Meaning
Shape	(119877, 27)	119,877 valid booking rows loaded into Pandas.
Missing supplier_id	240	Missing supplier was retained and flagged rather than treated as a fatal booking error.
Booking status	Completed 100,888; Cancelled 13,959; Confirmed 5,030	Status counts sum to 119,877.

3. Descriptive EDA and Booking Value Distribution

Before building charts, the analysis checks basic shape, missing values, descriptive statistics, and categorical distributions. This verifies that the SQL layer behaves as expected inside Python.

```
bookings['gross_booking_value_inr'].describe()
bookings['booking_status'].value_counts()
bookings.isna().sum().sort_values(ascending=False).head(10)
```

Metric	Verified value	Interpretation
Mean GBV	INR 218,154.25	Average booking value is relatively high because this is B2B travel.
Median GBV	INR 137,587.35	Median below mean indicates right-skew from high-value bookings.
Maximum GBV	INR 6,814,910.88	Very large bookings exist and require context, not automatic deletion.
Completed bookings	100,888	Majority of bookings are completed.
Cancelled bookings	13,959	Used for the booking cancellation KPI.

Metric	Verified value	Interpretation
Confirmed bookings	5,030	Open/confirmed bookings remain part of the booking population.

4. Outlier Analysis with the IQR Method

The Interquartile Range (IQR) method was used to flag unusually high or low GBV values. The method is a screening tool - it does not prove that a record is wrong.

```
q1 = bookings['gross_booking_value_inr'].quantile(0.25)
q3 = bookings['gross_booking_value_inr'].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
outliers = bookings[(bookings['gross_booking_value_inr'] < lower) |
                    (bookings['gross_booking_value_inr'] > upper)]
```

IQR component	Verified value
Q1	INR 70,939.81
Q3	INR 261,996.09
IQR	INR 191,056.28
Lower bound	INR -215,644.61
Upper bound	INR 548,580.51
Potential GBV outliers	9,207 bookings (7.68%)

Why the outliers were retained

Group	Average rooms	Average nights
Normal bookings	about 1.40	about 3.52
Potential GBV outliers	about 2.55	about 7.93

Interpretation: High-value bookings also tend to contain more rooms and longer stays. Many therefore look commercially plausible rather than like obvious data errors. Outlier does not mean invalid record.

5. Monthly Trend and MoM Growth

Monthly GBV was aggregated by booking month. Month-over-month growth compares each month with its immediately preceding month.

```
monthly = (bookings
    .groupby('booking_month', as_index=False)
    .agg(total_bookings=('booking_id','count'),
         gbv_inr=('gross_booking_value_inr','sum'),
         net_revenue_inr=('net_revenue_inr','sum')))
monthly['gbv_mom_pct'] = monthly['gbv_inr'].pct_change() * 100
```

Finding	Month	Verified result	Business meaning
Strongest GBV MoM growth	Dec 2024	+11.31%	Strong acceleration; investigate seasonality and booking mix.

Finding	Month	Verified result	Business meaning
Weakest GBV MoM growth	Feb 2024	-8.68%	Meaningful contraction; investigate demand and mix changes.
Coverage	Jan 2024-Dec 2025	24 months	Full booking period used in trend analysis.

Analyst caution: A trend identifies what changed, not necessarily why. Seasonality, promotions, supplier availability, or mix are hypotheses to investigate - not proven causes.

6. Agent, Destination, Supplier, and Hotel Analysis

Grouping by commercial entity helps distinguish volume leadership, revenue leadership, and margin efficiency. These are different concepts, so a single entity is not automatically "best" on every KPI.

Area	Verified finding	How to interpret it
Agents	7,995 active agents	Most agent IDs appear in the valid booking fact.
Top agent by bookings	A00145 - 249 bookings	Highest observed booking frequency.
Top agent by net revenue	A00238 - INR 6,823,003.54	Strongest observed agent net-revenue contribution.
Top destination	D0055 by GBV and net revenue	Strong on both transaction scale and revenue contribution.
High-cancel destination	D0119 - 18.84% among destinations with >=100 bookings	Useful target for operational investigation.
Top supplier	S0079 by GBV and net revenue	Largest supplier commercial contribution.
Highest supplier margin	S0052 - 13.05%	Strong margin efficiency among suppliers with meaningful volume.
Lowest supplier margin	S0143 - 11.67%	Candidate for cost/discount/contract review.
Top hotel by GBV	H00978	Highest hotel booking-value contribution.
Top hotel by net revenue	H00516	Highest hotel net-revenue contribution.
Highest hotel margin	H02897 - 15.42%	Strongest margin among hotels with >=50 bookings.
Lowest hotel margin	H02297 - 9.01%	Candidate for profitability investigation.

7. Cancellation, Refund, and Correlation Analysis

Cancellation analysis used the FX-normalized cancellation fact table. It contains 13,972 valid cancellation-event rows linked to valid bookings.

Cancellation insight	Verified result
Most common reason	Customer Change Of Plan - 4,823 events
Highest total refund reason	Customer Change Of Plan - about INR 1.021B
Highest average refund reason	Price Found Elsewhere - about INR 212,966.51

Cancellation insight	Verified result
Highest total cancellation fee reason	Customer Change Of Plan - about INR 36.32M

Correlation diagnostics

Relationship with GBV	Correlation	Interpretation
Rooms	0.460	Moderate positive association.
Nights	0.580	Moderate positive association; longer stays often have higher value.
Net Revenue	0.889	Strong positive association, expected because revenue is derived from booking economics.
Lead Days	-0.004	Almost no linear relationship in this dataset.

Correlation rule: Correlation does not prove causation. Use it as evidence of association and as a guide for deeper investigation.

8. Exports, Validation, Common Mistakes, and Interview Revision

Analysis outputs

The Python phase produced reusable analysis tables such as monthly_trend, agent_analysis, destination_analysis, supplier_analysis, hotel_analysis, and cancellation_summary for dashboarding and documentation.

Common mistakes and fixes

Mistake	Why it is a problem	Correct approach
Using pd.read_sql(query, sql) when sql is undefined	Raises a NameError.	Pass the SQLAlchemy engine/connection object.
Deleting every IQR outlier	Can remove legitimate high-value B2B transactions.	Investigate booking context first.
Calling correlation causal	Overstates what the data proves.	Use association language and propose follow-up analysis.
Exposing DB password	Security and portfolio risk.	Use environment variables or placeholders.

Interview-ready answer

60-second summary: I used Pandas on top of the cleaned PostgreSQL analytical layer to profile booking distributions, investigate potential GBV outliers, analyze monthly growth and partner performance, and study cancellation/refund drivers. I retained high-value IQR outliers after finding that they were associated with more rooms and longer stays, indicating plausible B2B behavior. The analysis identified Dec 2024 as the strongest GBV MoM month, D0055 and S0079 as major commercial contributors, and Customer Change Of Plan as the largest cancellation/refund driver.

Phase 6 completion checklist

- PostgreSQL connection verified and 119,877 valid booking rows loaded.
- EDA and IQR outlier investigation completed.
- Monthly trend and partner analyses completed.
- Cancellation/refund and correlation analyses completed.
- Reusable analysis outputs prepared for dashboarding/documentation.